

Module 6 Deep Learning

6.1 Module Overview									
Module Number	M6	Module Title	Deep Learning						
Stage of Principal Programme			Award	Semester	2	Duration. (Weeks F/T)	12 weeks	ECTS	10
Mandatory / Elective (M/E)	M	Hours of Learner Effort / Week	FT 4 PT 3						
Analysis of required hours of learning effort									
Teaching and Learning Modalities					✓if relevant to this module		Approx. proportion of total (hours)		
Contact Hours									
On-site face-to-face					✓		36 FT 3 PT		
Synchronous online*					✓		12 FT 33 PT		
Synchronous Hybrid									
Indirect/Non-Contact Hours									
Work Based									
Other (Independent learning, directed e-learning, assignments, asynchronous etc)							202 FT 214 PT		
Total							250		

**The hours for synchronous delivery should be read as “up to” with the proviso that full contact hours will be delivered. Any hours not covered by synchronous delivery will be covered by in-person face-to-face contact hours.*

Pre-Requisite Module, if any. Module # and Title		No
Co-Requisite Module, if any. Module # and Title		No
Maximum number of learners per instance of the module		120
Specification of the qualifications (academic, pedagogical and professional/occupational) and experience required of staff working in this module.		
Role e.g., Tutor, Mentor, Lecturer, Research Supervisor, etc.	Qualifications & experience required	Staff - Learner Ratio X:Y
Lecturer	Lecturing staff will have a minimum of a level 9 postgraduate diploma or master's degree in a relevant discipline with relevant and significant industry experience and/or PhD. Holders of level 8 honours degrees in a relevant discipline, who are exceptionally qualified by virtue of significant industry experience will also be considered.	1:30

Assessment Techniques – percentage contribution					
Continuous Assessment	100%	Proctored Exam – in person		Practical Skills Based	

Project		Proctored Exam - online		Work Based	
Capstone modules (Yes/No)?	No	If Yes, provide details			

6.2 Minimum Intended Module Learning Outcomes (MIMLOs)

MIMLO On completion of this module a learner will be able to:		Related MIPLO #
MIMLO 6.1	Research the structure and functions of the human neural system, including an introduction to artificial neural networks (ANNs), comparing biological and artificial neural network models.	MIPLO1
MIMLO 6.2	Evaluate ML (Machine learning) and DL (Deep learning) over the scalability of performance and need of data size. Understand the component of ANN in detail such as weight, bias, activation functions (sigmoid, tahn etc.) and features selections methods in case of textual data and numerical data.	MIPLO2 MIPLO8
MIMLO 6.3	Critically appraise and apply a comprehensive understanding of feed-forward networks, including loss functions, optimization, gradient descent, backpropagation, and learning rates, to design and optimize neural network models.	MIPLO3 MIPLO7
MIMLO 6.4	Evaluate and apply various evaluation techniques of DL (Classification and Regression) such as log loss, precision, recall, absolute error etc. Study the significance of evaluation techniques based on specific problems.	MIPLO4 MIPLO5 MIPLO9
MIMLO 6.5	Critically evaluate various neural network architectures (e.g., feedforward networks, convolutional neural networks, recurrent neural networks). Understand feature engineering approaches based on the chosen neural network architecture.	MIPLO6 MIPLO10
MIMLO 6.6	Critically design neural network architectures for specific applications, justifying the selection of parameters used in Deep Learning software libraries.	MIPLO2 MIPLO4 MIPLO5

6.3 Indicative Module Content, Organisation and Structure

Week 1

Introduction to Neural Networks

- Fundamentals of Neural Networks
- Multi-Layer Perceptron (MLP)
- Introduction to Activation Functions and Gradient Descent

Week 2

Artificial Neural Networks (ANN)

- Structure and functionality of ANN

- Training and Optimization techniques
- Application examples and practical implementation

Week 3**Convolutional Neural Networks (CNN) Basics**

- Overview of CNN architecture (convolution, pooling, fully connected layers)
- Introduction to applications in image processing and computer vision

Week 4**Advanced CNN Architectures**

- Extensions and variations (e.g., ResNet, Inception, VGG)
- Transfer Learning and Fine-Tuning with CNNs

Week 5**Recurrent Neural Networks (RNNs)**

- Basic structure and use cases for sequence data
- Practical implementation of RNNs in text and time-series data

Week 6**Advanced RNNs – LSTM and GRU**

- Long Short-Term Memory (LSTM) Networks and GRUs
- Use cases, strengths, and limitations for language modeling and sequential data

Week 7**Stacked Auto-Encoders**

- Concept of auto-encoders and stacked architectures
- Use in data compression, denoising, and feature extraction

Week 8**Deep Belief Networks (DBN) and Deep Boltzmann Machine (DBM)**

- Understanding DBN and DBM architectures
- Application in feature learning and representation learning

Week 9**Generative AI and Transformer Architectures**

- Generative AI overview
- Transformer architectures: attention mechanisms and encoder-decoder structures

Week 10**ChatGPT and BERT**

- ChatGPT and BERT: structure, use cases, and fine-tuning for NLP tasks

- Hands-on with transformer-based language models

Week 11**Course Review and Emerging Trends**

- Recap of all covered neural networks and models
- Emerging trends in Gen AI, ethical considerations, and real-world applications

Week 12

Presentation and Demonstration

6.4 Work-based learning and practice-placement (if applicable)

N/A

6.5 Specific module resources required (if applicable)

Lecture room fitted with electronic whiteboard, lecturer computer and overhead projector, learner computers configured with technical resources and software as appropriate. See program document section 8 for detail regarding physical resources.

6.6 Application of programme teaching, learning and assessment strategies to this module

All modules have a learning space on DBS's learning management system (LMS), Moodle, which is used as a blended delivery resource for learners. This system provides a single point of entry for learners to access digital resources to enhance and support the learning experience. Links to core academic materials, continuous assessments, readings, videos and reusable learning objects (RLOs) and other directed e-learning activities are made available on Moodle.

The content allows the learner to progress through the concepts and tools in an orderly and logical fashion. The assessment strategy is planned to allow for submission on an ongoing basis thus ensuring the progressive building of skills by the learner early in the module, which facilitates feedback and underpins the overall learning experience. The learners will have directed e-learning and progressive practical skills-based exercises to complete individually during the semester. The lecturer will provide timely feedback for the exercises to guide learners to successfully progress and complete their summative assessments. Exercises will provide a review of progress of learners and problems will be rectified at early stages.

Students first encounter the basic building blocks, like the multi-layer perceptron, activation functions, and the crucial concept of gradient descent. With this foundation in place, they delve deeper into the structure and inner workings of artificial neural networks, learning how to train and optimize these powerful models for various tasks.

The module then takes them through the intricacies of convolutional neural networks (CNNs), where they uncover the magic behind image recognition and computer vision. Starting with the core architectural components – convolution, pooling, and fully connected layers – they gradually explore more sophisticated CNN architectures like ResNet, Inception, and VGG. Along the way, they discover the power of transfer learning, allowing them to leverage pre-trained models and adapt them to new challenges.

Next, the module ventures into the realm of sequential data, where recurrent neural networks (RNNs) reign supreme. Students grasp the fundamental structure of RNNs and witness their prowess in handling text and time series data. This exploration leads them to advanced RNN architectures like LSTMs and GRUs, unraveling their ability to capture long-term dependencies in sequential information.

As the module progresses, the spotlight turns to the exciting domain of generative AI and the revolutionary transformer architectures. Students are introduced to the core concepts of generative AI and its diverse applications. They then delve into the intricacies of transformers, unraveling the mechanisms of attention and the elegant encoder-decoder structure that lies at the heart of many cutting-edge language models.

The exploration culminates with a deep dive into ChatGPT and BERT, two prominent transformer-based language models that have revolutionized natural language processing. Students gain hands-on experience with these models, exploring their capabilities and fine-tuning them for specific NLP tasks.

Throughout this immersive learning experience, students are supported by a blended learning approach. The DBS learning management system (Moodle) serves as a central hub for accessing digital resources, including core academic materials, continuous assessments, readings, videos, and reusable learning objects. This rich learning environment fosters continuous learning and skill development.

6.7 Summative Assessment Strategy for this module

<i>MIMLOs</i>	<i>Technique(s)</i>	<i>Weighting</i>
1,2,4	CA01	40%
3,5	CA02	60%

6.8 Sample Assessment Materials

Assessment 1:

Item	Description
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Module Title:	Deep Learning
Module Code:	B9AI104
Module Lecturer (s):	Dr Shahram Azizi Sazi
Programme/Cohort:	Msc in AI
Method of Assessment:	
Percentage (%) Weighting:	40
MIMLOs being assessed:	
Assessment Number:	1
Individual/Group:	Individual
Issue Date:	
Submission Date:	Week 8
Feedback Date:	Weeks 9-12
Feedback Strategy:	In-class

This assignment is an individual project, and each learner is required to submit their final report on Moodle. In addition to the main report, each learner must submit a Word document that includes notes on the project development process and a detailed account of their contribution, specifying the precise nature of their work and estimated time of completion. This document will represent the learner's individual work, and adjustments to grades may be applied as necessary.

Report Requirements:

The report should include the following sections:

- **Model Architecture Background:** Provide an overview of the architecture for three selected Deep Learning models, detailing the scope, types of models, and mathematical foundations. Additionally, include a list of rules for each model. **(20 marks)**
- **Dataset Implementation:** Demonstrate the dataset implementation, ensuring it is error-free. **(30 marks)**
- **Deep Learning Design:** Discuss the design of each Deep Learning model, highlighting strengths, weaknesses, and notable artifacts. Include conclusions, a bibliography, and relevant screenshots with explanations where required. **(20 marks)**
- **Application Deployment:** Implement and deploy insights in a micro-framework, ideally as a web application. **(30 marks)**

Presentation and Demonstration:

A presentation and live demonstration of the project are mandatory components of this assignment. Each learner will be required to showcase their model's design, implementation, and functionality to demonstrate a thorough understanding and application of deep learning concepts.

Assignment Objectives:

1. Analyze the requirements in alignment with organizational goals and present an individual deep learning model.
2. Design an individual deep learning model.
3. Implement a deep learning algorithm for a practical, real-world problem.

4. Present and demonstrate the deep learning model's functionality and insights.

Resources

Software or tools required/ useful: GoogleColab, Vscod, Render.com

Templates, files, data tables provided by the lecturer:

<https://colab.research.google.com/drive/1xuz2Pm55jPOzyDu275XVqTrGDJmhgy3k#scrollTo=f-RGi9PMqRMz>

Generative Artificial Intelligence Assessment Scale

Can generative AI be utilised in this assignment?

4
AI TASK COMPLETION, HUMAN EVALUATION
AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output.
You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).

- a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
- a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.
 - e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.

6.9 Indicative reading lists and other information resources

Primary Text:

Wirsansky, E (2024) Hands-on genetic algorithms with Python : apply genetic algorithms to solve real-world AI and machine learning problems, Birmingham Packt Publishing

Pedrycz, W. and Chen, S.M. (2020) *Deep Learning: Concepts and Architectures*. Springer.

Supplementary Reading:

Moolayil, J. (2019) *Learn Keras for Deep Neural Networks. A Fast-Track Approach to Modern Deep Learning with Python*. Apress.

Literature Resources

1. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
 - o A foundational text covering MLP, CNNs, RNNs, LSTMs, autoencoders, and probabilistic models.
 - o Provides theoretical and mathematical explanations suitable for intermediate to advanced learners.
2. "Deep Belief Nets" by Geoffrey Hinton (Journal Papers)
 - o Hinton's original research on Deep Belief Networks and Boltzmann Machines.
 - o Accessible through journals like *Neural Computation* or *Journal of Machine Learning Research*.
3. "Attention is All You Need" by Vaswani et al.
 - o The seminal paper introducing the Transformer model, foundational for understanding Generative AI, BERT, and ChatGPT.
 - o Published in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

fast.ai - Practical Deep Learning for Coders

- Provides a practical, code-focused deep learning course with an emphasis on high-level PyTorch applications.

Hugging Face. (n.d.). *Hugging Face Model Hub and Transformers Library*. Retrieved from <https://huggingface.co/>